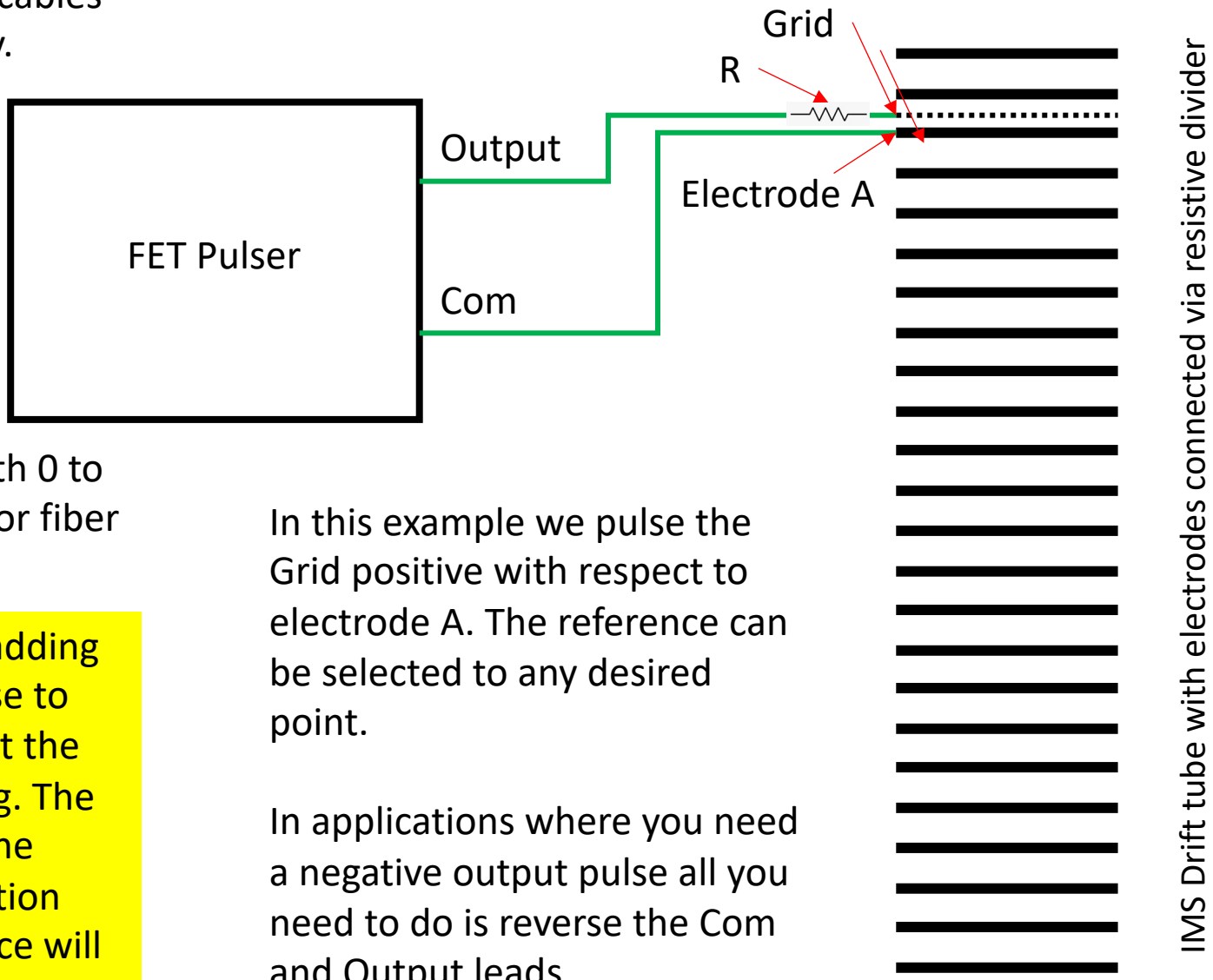
FET Pulser R6.0

GAA Custom Electronics, LLC

July 23, 2022
Revised October 19, 2022
Revised December 16, 2023
Revised August 23, 2024

Typical application

Use the supplied cables and power supply.



Control pulser with 0 to 5V control signal or fiber cable.

We recommend adding the resistor R close to the grid to protect the pulser from arcing. The larger the value the better the protection but large resistance will slow the rise and fall time. Try 100K to 10K.

In this example we pulse the Grid positive with respect to electrode A. The reference can be selected to any desired point.

In applications where you need a negative output pulse all you need to do is reverse the Com and Output leads.

Overview

- High speed FET Pulser.
  - Pulse output relative to common
  - Rise and fall times ~15nS
  - Adjustable pulse voltage from 20 to 200 volts
  - 5 watt maximum power dissipation
  - Floatable to 3KV, 10KV using optional isolator
  - 0 to 5 volt input control voltage, high Z input or fiber optic trigger cable
  - Powered from 12V wall supply, 125mA nominal current draw
  - Overvoltage protection using 300 volt GDTs
  - Small size, 2.05” x 2.85”, about 0.6” of total height
  - 50M ohms connecting the pulser float ground to the control side of pulser. This is needed to support the 12VDC isolator module.

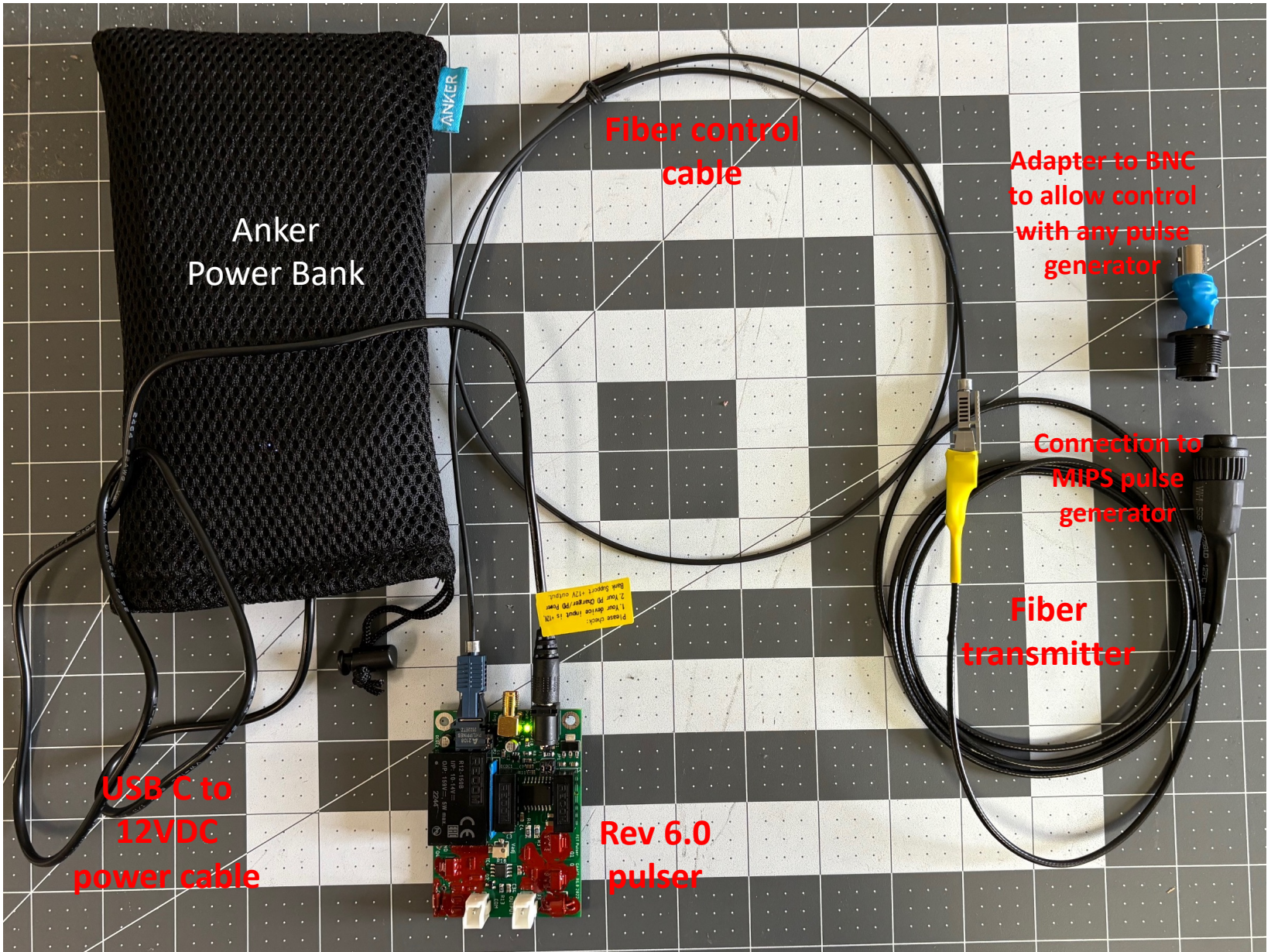
Non isolator pulse option

- The pulser is available without the optional isolation module for application where 3KV of isolation provided by the standalone pulser is adequate.
- In this configuration the fiber optic input is not installed.
- The non isolator version kit includes the pulser, connection cables, a 12V wall supply and a SMA to BNC cable.
- You will notice red HV varnish we applied to the pulser’s high voltage area. This was done to provide additional protection, you should still avoid contact with the pulser when its is powered.

High voltage isolation

- The pulser can be powered using a USB C power bank, such as a Anker 537, 65W model. This will easily power the a pulser for several days.
- A USB C to DC 2.5mm x 2.5nn 12V power cord will allow connecting the bank to the pulser. Plug the cable into the pulser first then the power bank.
- Make sure the power bank and the power cable are isolated from ground.
- Using this configuration and the fiber signal interface will allow you to float the pulser at high voltages, in excess of 20KV.

High voltage isolation, connections



Typical Pulser performance data

Delay from control to output = ~60nS

Rise and fall time = ~10nS



Blue = Control signal, Yellow = Pulser output



Pulser Connection Diagram

+12VDC power
Center positive

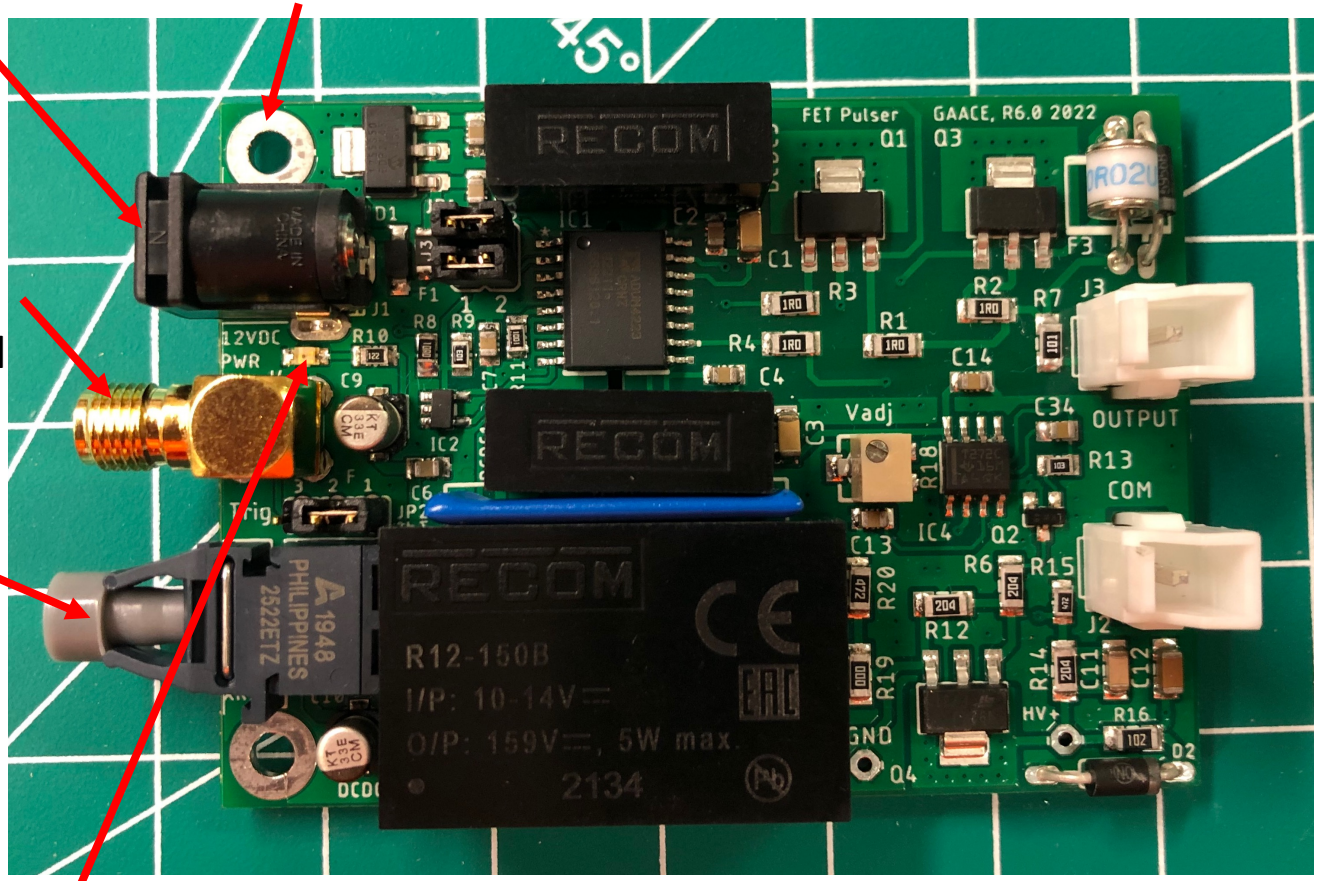
Mounting holes connected to ground

SMA input
0 to 5V control

Fiber input
trigger

Pulse output

Common



Power on green LED

Ground referenced

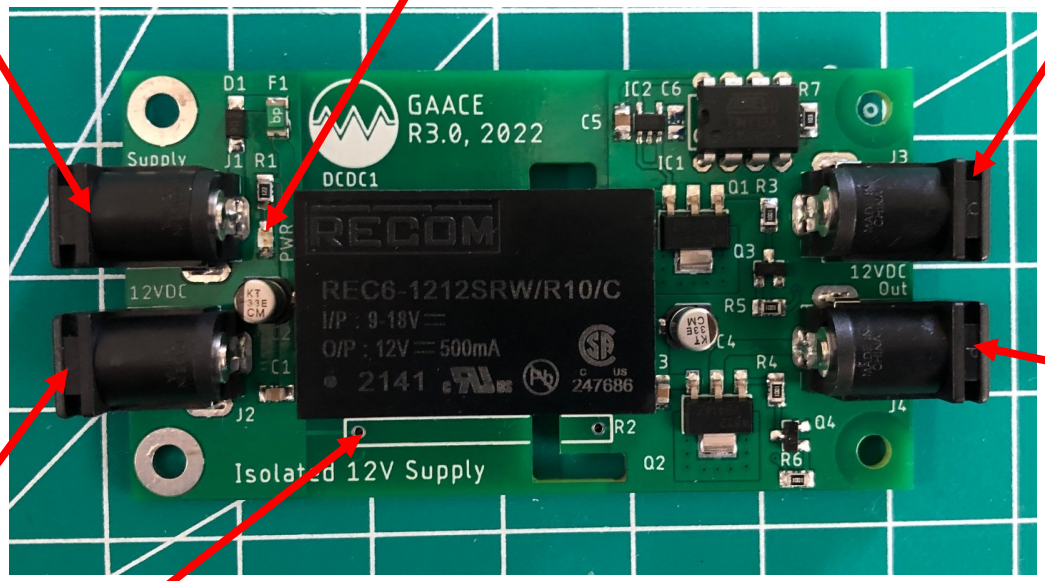
Floating

12VDC Isolation module

Optional, Only used when floating to 10KV

12VDC input from wall supply. Only use the wall supply provided with the grounded power cable. Two connectors are provided, they are internally connected. You can use one to apply power and then other connector to send power to a second isolator if needed

Power on green LED



This module is a 12VDC in to 12VDC out isolator designed to increase the pulsers isolation voltage.

Optional isolation resistor, please see isolation discussion.

12VDC isolated output. This output will connect to the power input of the pulser. It is possible to power two pulsers from one isolator. The two outputs are sequenced on with a delay of a few seconds between each output.

Note! The output section of this isolator will be at the float voltage of your system!!

Pulser Kit Components

Optional isolation module shown

12VDC wall adapter. Use only the wall adapter provided and make sure you use a grounded 3 prong power cable.

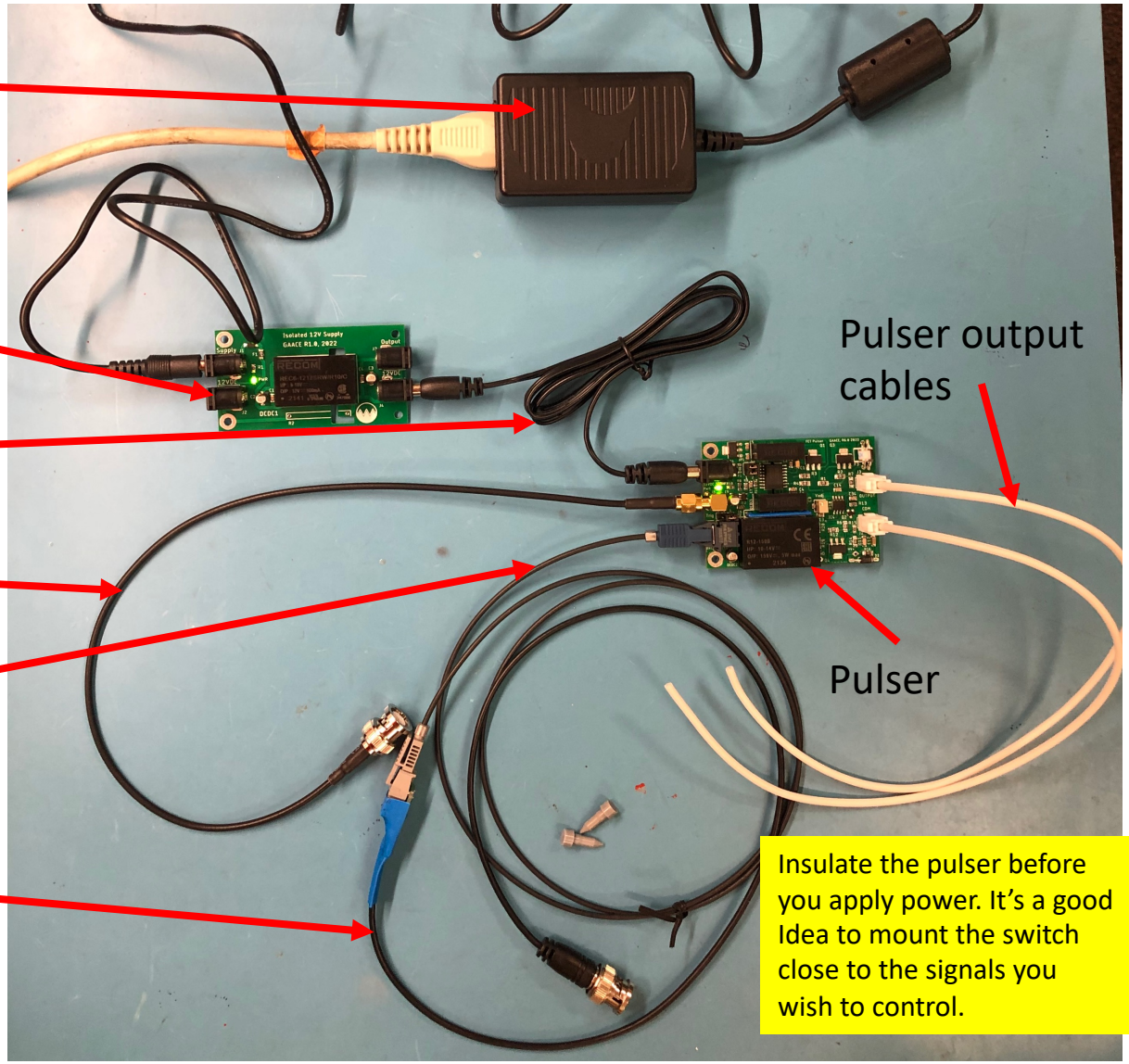
12VDC isolation module

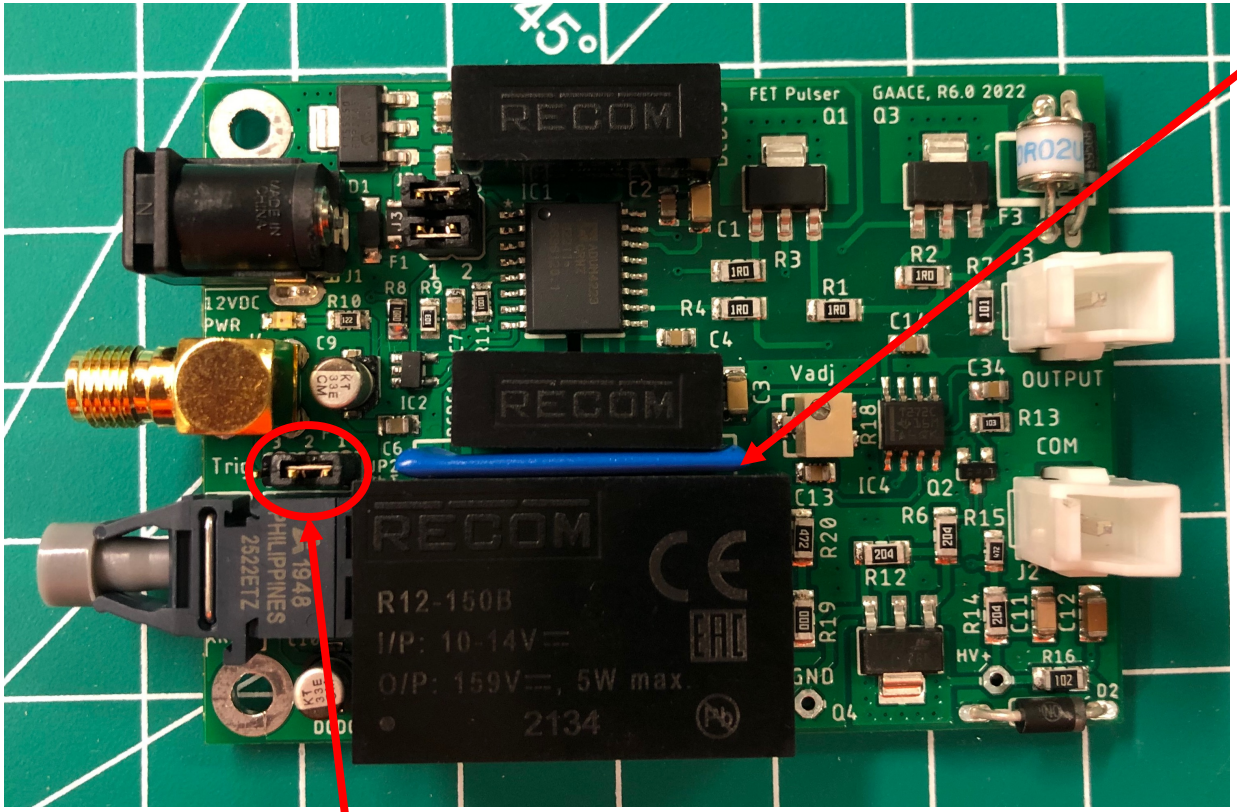
12VDC power cable

SMA to BNC trigger cable

Fiber trigger cable

Fiber transmitter

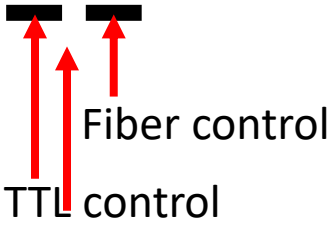


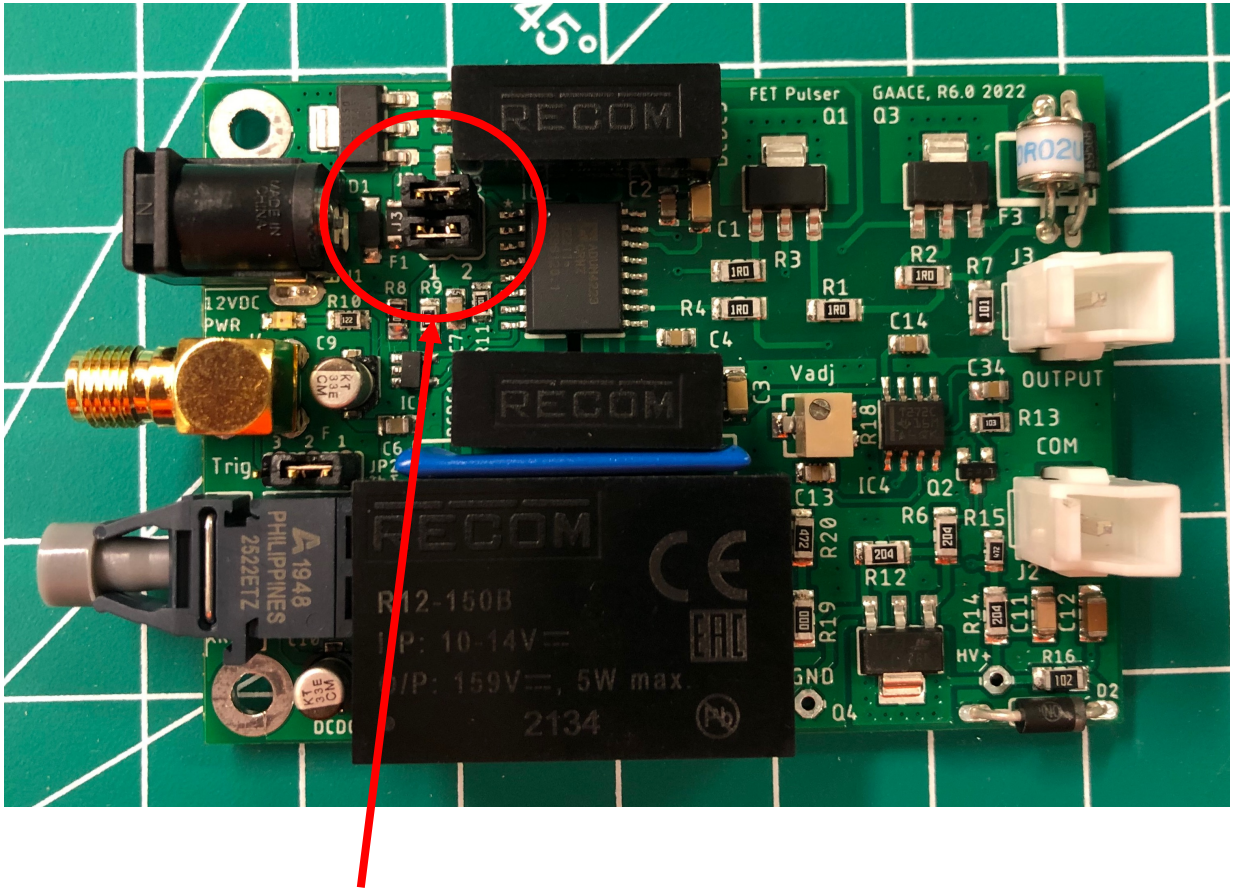


This resistor is provided but not installed. Please see the discussion of isolation to describe isolation options.

This three pin jumper block allows you to select the control mode for the pulser. When the shorting jumper is install in the left most position then the TTL trigger signal applied using the SMA connector will trigger the pulser. When the jumper is in the right most position then the fiber optic input will control the pulser.

Control selection options





These jumpers allow you to change the logic level for the pulser. When positioned as shown a logic 1 (5 volts on the SMA connector) will turn the pulser on. When positioned with the jumpers horizontal a logic 0 (0 volts) will turn the pulser on . **Note! When using the fiber option cable and transmitter these logic levels are reversed.**



Logic 1 = ON

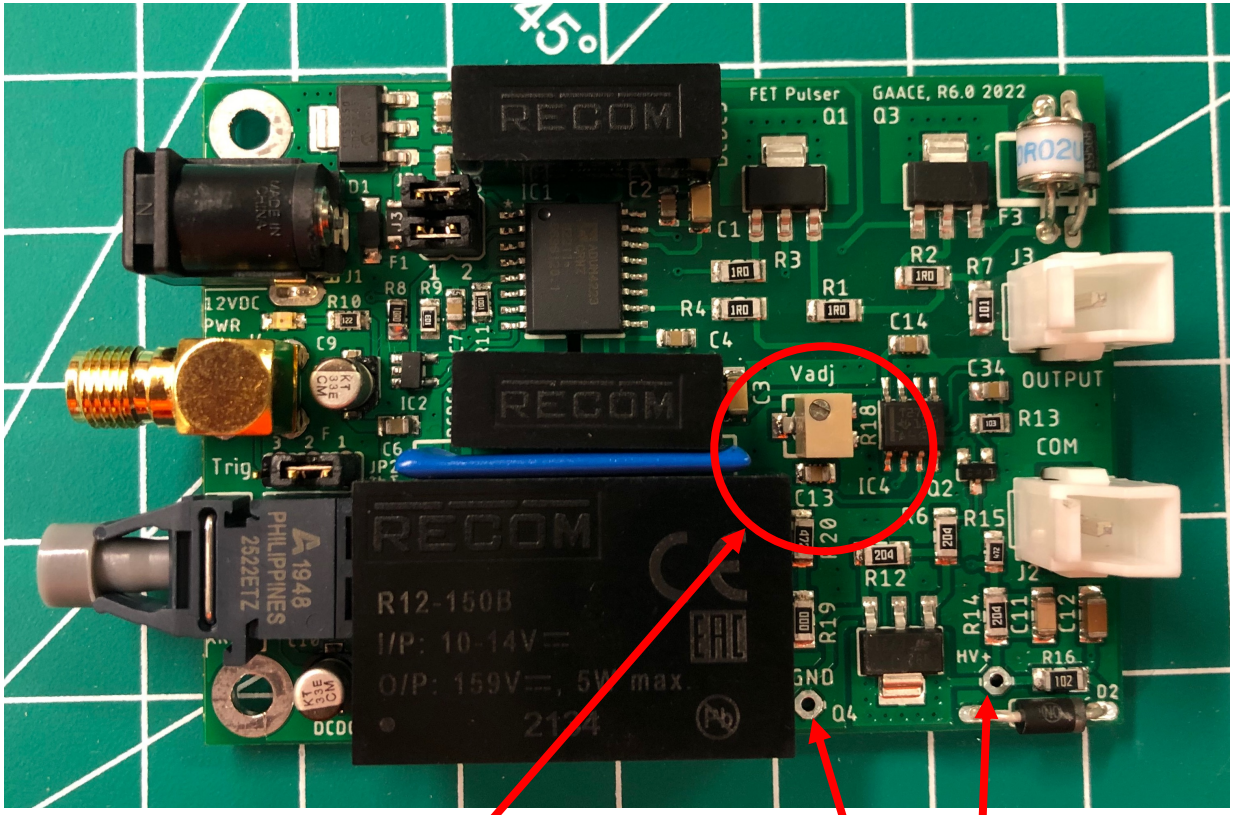


Logic 0 = ON

Setting the Pulser voltage

Make all adjustments and measurements with float voltage off!

Use the supplied cable to apply power. You do not need a control pulse to set the voltage.



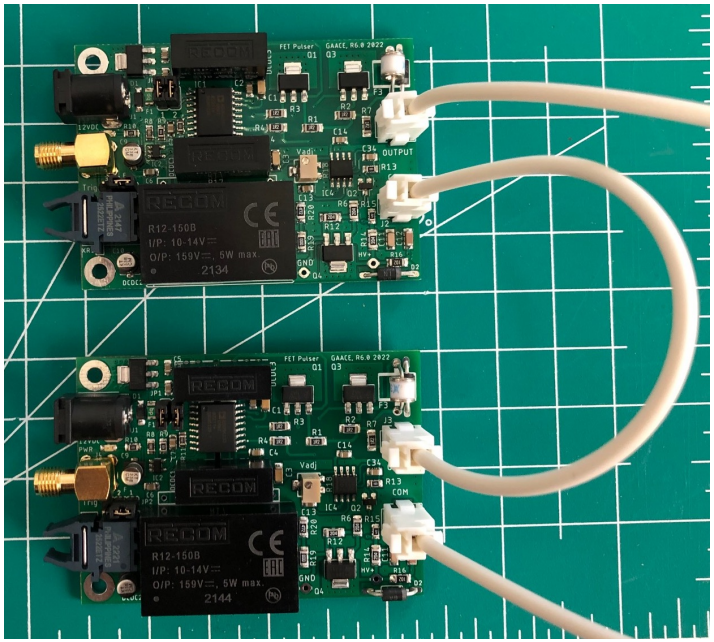
Note: Make this adjustment with the output and com cables disconnected.

Adjust this potentiometer to reach the desired pulse output. Do not exceed 200V

Measure the voltage between these two points with a multimeter, you can also set the pulse voltage by using a scope to monitor the output pulse.

Stacking multiple pulsers

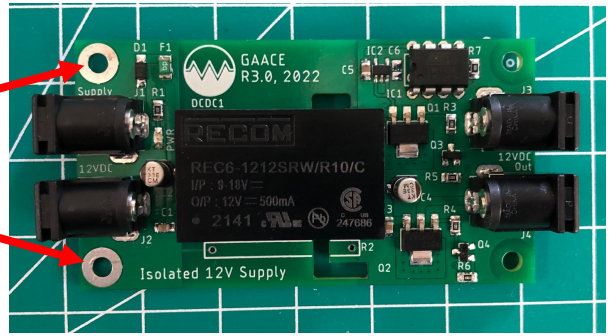
Multiple pulser can be stacked as shown in this picture. This will allow you to generate multistate output pulse sequences.



Isolation (optional)

- The pulser has 3KV of isolation from its outputs to the control inputs.
- The isolation module provides 10KV of isolation.
- We include leakage resistors with your pulsers, these are 50M resistors that connect the isolated output to the control side ground. In applications where you are using the isolation module and fiber cable triggering these resistors will make sure that the control size of the pulser is not floating, it will be referenced to the float voltage.
- The isolation module in powered with a 12VDC wall supply with the ground connected to the wall supply ground. We recommend that you bond the mounting lugs on the isolation module to earth ground.

Bond to ground

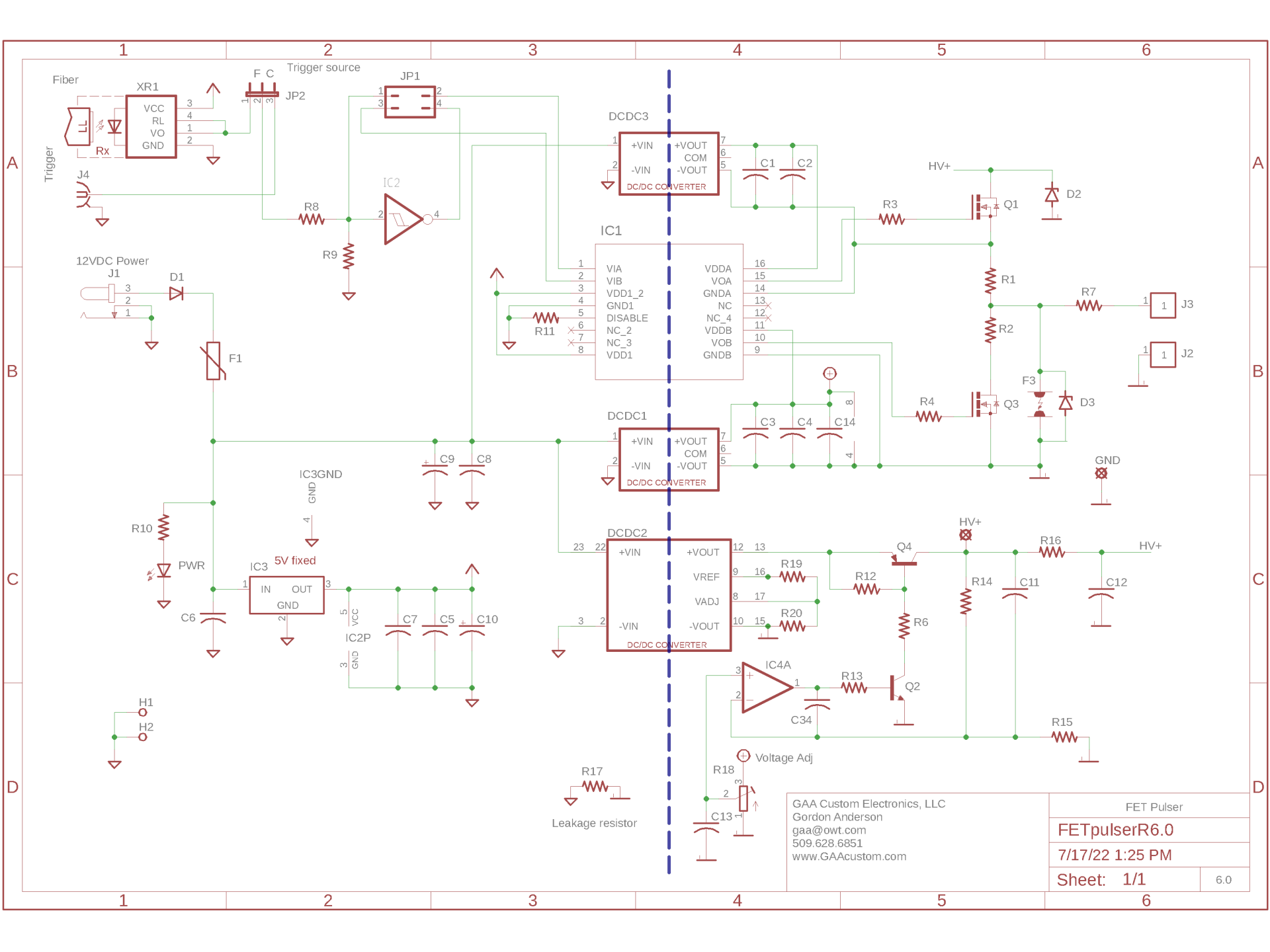


Powering up the system

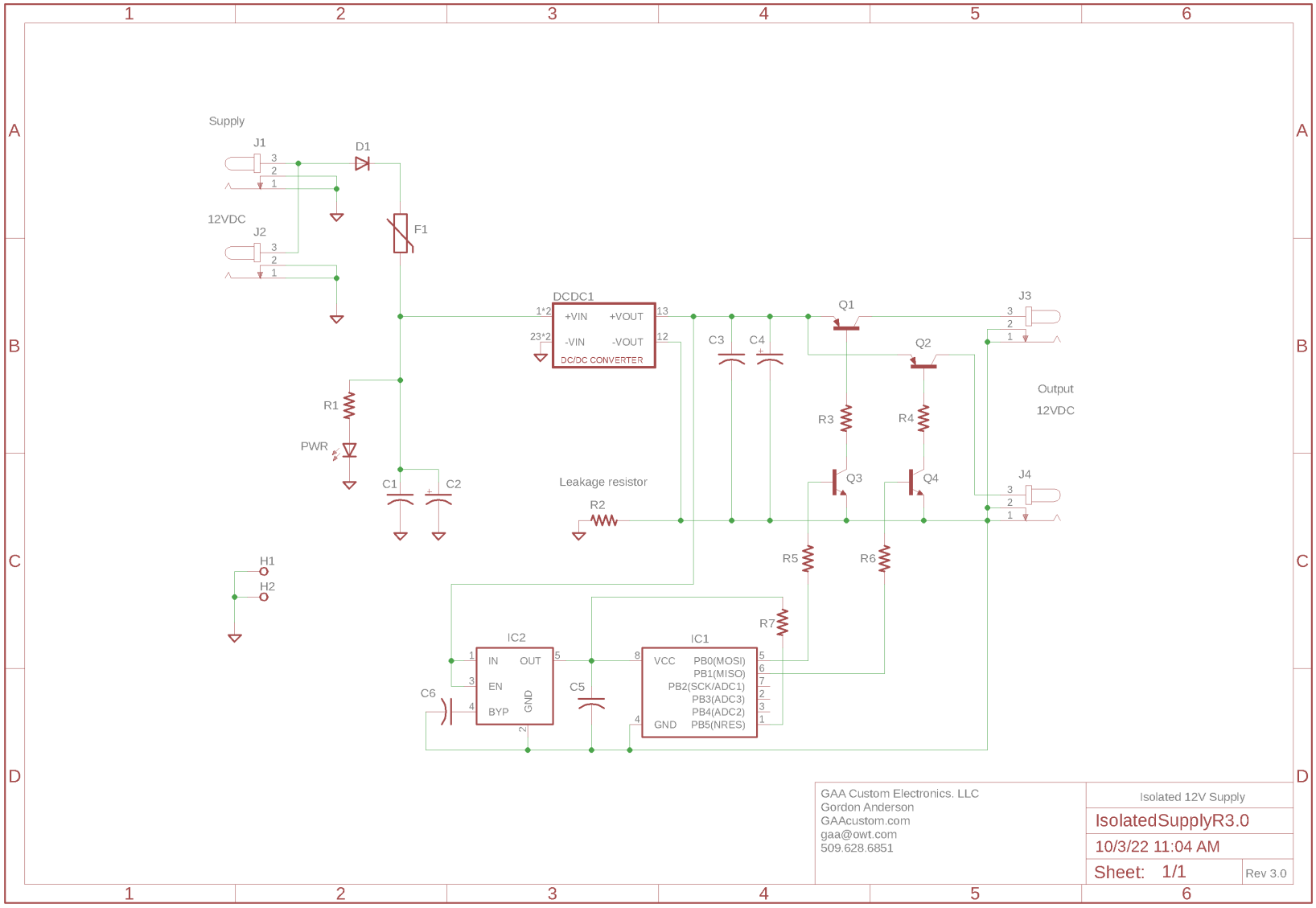
- Depending on your application and power configuration there are a number of ways power can be applied to the pulser.
- Remember there is no power switch so when you plug power into the pulser it will turn on and 200VDC will be present on the pulser PCB.
- The pulser and the isolator have green power on LEDs to let you know they are powered up.
- When powering the pulser using the isolator you will notice a delay is the output power. This is done to sequence on the load and allow running two pulsers from one isolator.
- It is possible to run two pulsers from one isolation module as long as the total power requirement does not exceed the isolator limit.

Power

- The pulser is capable of dissipating 5 watts of power.
- Generally these pulsers are using in applications that drive electrodes that present a capacitive load.
- Power can be estimated knowing the load capacitance, pulse voltage, and frequency.
- The key equations you need are:
  - Energy in a capacitor in joules J = 1/2CV^2 , C = capacitance in farads and V = voltage in volts
  - A joule per second is a watt.
  - The pulser output voltage in volts.
  - If you drive the pulser with a 10KHz square wave that is actually 20,000 charge discharge events in one second.
  - Note that the power goes up with the square of voltage!


Fiber
XR1
VCC
RL
VO
GND
Rx
Trigger
J4
F C
JP2
Trigger source
JP1
DCDC3
+VIN
-VIN
+VOUT
COM
-VOUT
DC/DC CONVERTER
C1
C2
IC2
R8
R9
IC1
VIA
VIB
VDD1_2
GND1
DISABLE
NC_2
NC_3
VDD1
VDDA
VOA
GNDA
NC
NC_4
VDDB
VOB
GNDB
R11
HV+
Q1
D2
R3
R1
R2
R7
J3
J2
F3
D3
Q3
R4
12VDC Power
J1
D1
F1
DCDC1
C3
C4
C14
C9
C8
GND
IC3GND
R10
PWR
C6
IC3
5V fixed
IN
OUT
GND
IC2P
VCC
C7
C5
C10
DCDC2
+VIN
-VIN
+VOUT
VREF
VADJ
-VOUT
DC/DC CONVERTER
R19
R20
R12
Q4
R6
R14
C11
R16
C12
HV+
IC4A
R13
Q2
C34
R15
Voltage Adj
R18
C13
H1
H2
R17
Leakage resistor
GAA Custom Electronics, LLC
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www.GAAcustom.com
FET Pulser
FETpulserR6.0
7/17/22 1:25 PM
Sheet: 1/1
6.0

FET pulser, R6.0, BOM									
Qty	Value	Device	Package	Part	Description	SUPPLIER	Part Number	Cost Each	Cost Total
2	10uF, 35V	C-USC1206	C1206	C1, C3	CAP CER 10UF 35V X6S 1206	Digikey	490-12425-1-ND	$0.47	$0.94
1	1000pF, 1KV	C-USC1206	C1206	C11	1000pF, 1000V, 1206	Digikey	709-1029-1-ND	$0.20	$0.20
1	0.1uF, 450V	C-USC1206	C1206	C12	0.1uF, 450V, 1206	Digikey	445-7758-1-ND	$0.44	$0.44
9	0.1uF, 100V	C-USC0805	C0805	C2, C4, C5, C6, C7, C8, C13, C14, C34	CAP CER 0.1UF 100V X7R 0805	Digikey	1276-2447-2-ND	$0.28	$2.52
2	47uF, 16V	CPOL-USB	PANASONIC_B	C9, C10	CAP ALUM 47UF 20% 16V SMD	Digikey	P15090CT-ND	$0.57	$1.14
1		DIODE-SOD123	SOD123	D1	DIODE SCHOTTKY 60V 1A SOD123	Digikey	MBRX160TPMSCT-ND or SMD16PL-TPMSCT-ND	$0.45	$0.45
2		ZENER-DIODEDO35Z10	DO35Z10	D2, D3	DIODE ZENER 200V 3W AXIAL	Digikey	1N5956BGOS-ND	$0.46	$0.92
2	15V in, 12V out	RP-XX	RP-XX	DCDC1, DCDC3	DC DC CONVERTER 12V 1W	Digikey	945-1254-ND	$13.31	$26.62
1	R24-100B	R24-100B	RXX-B	DCDC2	DC/DC CONV INNOLINE ADJUSTABLE	Digikey	945-2053-5-ND	$42.43	$42.43
1	SPARKFUN-POWERIC_PTCSMD	SPARKFUN-POWERIC_PTCSMD	SPARKFUN-POWERIC_PTC-1206	F1	PTC RESET FUSE 30V 350MA 1206	Digikey	F8473TR-ND	$0.57	$0.57
1	300V	ARRESTER	EPCOS_N81	F3	GDT 300V 2.5KA 2 POLE TH	Digikey	GTCA25-301M-R02-ND	$2.90	$2.90
1	ADUM4223CRWZ-RLRW_16_ADI-M	ADUM4223CRWZ-RLRW_16_ADI-M	RW_16_ADI-M	IC1	DGTL ISO 5KV 2CH GATE DVR 16SOIC	Digikey	ADUM4223CRWZ-RLCT-ND	$6.65	$6.65
1	NC7SZ14M5X	NC7SZ14M5X	SOT23-5	IC2	TinyLogic(R) UHS Inverter with Schmitt Trigger Input	Digikey	NC7SZ14M5XCT-ND	$0.45	$0.45
1		LT1129CST5	SOT223	IC3	VOLTAGE REGULATOR, 5V	Digikey	296-12290-1-ND	$0.66	$0.66
1	OPA2725D	OPA2725D	SO08	IC4	IC OPAMP GP 3MHZ RRO 8SOI	Digikey	296-26806-1-ND	$1.04	$1.04
1		JACK-PLUG1	SPC4078	J1	CONN JACK POWER 2.0MM PCB	Digikey	CP-102A-ND	$1.00	$1.00
2	1-1123824-1	1-1123824-1	TE_1-1123824-1	J2, J3	CONN HEADER VERT 1POS	Digikey	A30708-ND	$0.38	$0.76
1	BU-SMA-V	BU-SMA-V	BU-SMA-V	J4	CONN SMA JACK R/A 50OHM PCB	Digikey	931-1361-ND	$3.30	$3.30
1		JP2Q	JP2Q	JP1	CONN HEADER VERT 4POS 2.54MM	Digikey	609-3201-ND	$0.38	$0.38
1		JP2E	JP2	JP2	CONN HEADER VERT 3POS 2.54MM	Digikey	609-3461-ND	$0.34	$0.34
1		LEDCHIP-LED0805	CHIP-LED0805	PWR	LED GREEN DIFFUSED 0805 SMD	Digikey	475-1410-1-ND	$0.25	$0.25
2	N-HEXFET-2D_SOT232-	N-HEXFET-2D_SOT232-	SOT223	Q1, Q3	MOSFET N-CH 600V 1.5A SOT223	Digikey	1016-2071-1-ND	$0.58	$1.16
1	NPN 500V	BC818-NPN-SOT23-BEC	SOT23-BEC	Q2	NPN Transistor, 500V, SOT23	Digikey	568-6770-1-ND	$0.47	$0.47
1	PNP 500V	FZT753SMD	SOT223	Q4	PNP Transistor, 500V, SOT223	Mouser	511-STN9260 or 497-12275-1-ND	$0.80	$0.80
5	1	R-US_R1206	R1206	R1, R2, R3, R4, R7	1 ohms, 1206	Digikey	P1.0ECT-ND	$0.10	$0.50
1	1200	R-US_R0805	R0805	R10	1200 ohms, 805	Digikey	P1.2KACT-ND	$0.10	$0.10
1	1000	R-US_R0805	R0805	R11	1000 ohms, 0805	Digikey	P1.0KACT-ND	$0.10	$0.10
1	4.7K	R-US_R0805	R0805	R15	RES SMD 4.7K OHM 5% 1/8W 0805	Digikey	P4.7KACT-ND	$0.10	$0.10
1	1K	R-US_R1206	R1206	R16	1K ohm, 1206	Digikey	1.0KECT-ND	$0.10	$0.10
1	50M	R-US_HB	HB_RES	R17	RES 50M OHM 1% 1W RADIAL	Digikey	A105540-ND	$9.36	$9.36
1	5K	R-TRIMM4G/J	RTRIM4G/J	R18	TRIMMER 5K OHM 0.25W GW TOP ADJ	Digikey	3214X-1-502ECT-ND	$3.04	$3.04
1	0	R-US_R1206	R1206	R19	RES SMD 0 OHM JUMPER 1/4W 1206	Digikey	P0.0ECT-ND	$0.10	$0.10
1	4700	R-US_R1206	R1206	R20	RES SMD 4.7K OHM 5% 1/4W 1206	Digikey	P4.7KECT-ND	$0.10	$0.10
3	200K	R-US_R1206	R1206	R6, R12, R14	RES SMD 200K OHM 5% 1/4W 1206	Digikey	P200KECT-ND	$0.15	$0.45
1	10	R-US_R0805	R0805	R8	RES SMD 10 OHM 5% 1/8W 0805	Digikey	P10ACT-ND	$0.10	$0.10
2	10K	R-US_R0805	R0805	R9, R13	RES SMD 10K OHM 5% 1/8W 0805	Digikey	P10KACT-ND	$0.10	$0.20
1	HFBR-2521	HFBR-2521	HFBR-15X3	XR1	FIBER OPTIC RCVR 1.67V VERSATILE	Digikey	516-2305-5-ND	$17.42	$17.42
					FIBER OPTIC TRANSMITTER 660NM	Digikey	516-2304-5-ND	$17.42	$0.00
					FIBER OPTIC CBL LATCH SIMPLEX 1M	Digikey	516-2088-ND	$8.23	$0.00


Supply
J1
D1
12VDC
J2
F1
DCDC1
+VIN
+VOUT
-VIN
-VOUT
DC/DC CONVERTER
C3
C4
Q1
Q2
J3
Output
12VDC
R1
PWR
R3
R4
C1
C2
Leakage resistor
R2
Q3
Q4
J4
H1
H2
R5
R6
R7
IC2
IN
OUT
EN
GND
BYP
C6
C5
IC1
VCC
PB0(MOSI)
PB1(MISO)
PB2(SCK/ADC1)
PB3(ADC3)
PB4(ADC2)
GND
PB5(NRES)
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509.628.6851
Isolated 12V Supply
IsolatedSupplyR3.0
10/3/22 11:04 AM
Sheet: 1/1
Rev 3.0

Isolated Power Supply, R2.0, BOM

Qty-1	Part References	Value	Package	Vendor Part Num	VID	Mfg Part Num	Description	Cost	Order Num	Num Ordered	Stock		XREF
4	C1, C3, C5, C6	0.1uF, 100V	C0805	1276-2447-2-ND	DK	CL21B104KCC5PNC	CAP CER 0.1UF 100V X7R 0805	$0.05	77200589.xlsx	40008000	1000		REF485
2	C2, C4	47uF, 16V	PANASONIC_B	P15090CT-ND	DK	EEE-FT1C470AR	CAP ALUM 47UF 20% 16V SMD	$0.20	60645765.xlsx	125	*		REF251
1	D1		SOD123	SMD16PL-TPMSCT-ND	DK	SMD16PL-TP	DIODE SCHOTTKY 60V 1A SOD123FL	$0.20	76937552.xlsx	150	150		REF1482
1	DCDC1	REC6-1212SRW/R10/C	REC6-XX/R10/C	919-6-1212SRWR10C	ME	REC6-1212SRW/R10/C	Isolated DC/DC Converters Isolated DC/DC Converters 6W DC/DC 10kV REG 12Vin 12Vout	$51.06	261535425.xls	23	*		REF1965
1	F1	SPARKFUN-POWERIC_PTCSMD	SPARKFUN-POWERIC_PTC-1206	507-1805-1-ND	DK	0ZCJ0075AF2E	PTC RESET FUSE 16V 750MA 1206	$0.19	76937552.xlsx	250	*		REF116
1	IC1	ATTINY13DIP	DIP08	ATTINY13A-PU-ND	DK	ATTINY13A-PU	IC MCU 8BIT 1KB FLASH 8DIP, Must be programmed before install	$1.22	77832583.xlsx	10	10		REF1548
1	IC2		SOT23-5	LT1761ES5-5#TRPBFCT-ND	DK	LT1761ES5-5#TRPBF	IC REG LINEAR 5V 100MA TSOT23-5	$2.87	NoOrderID.xlsx	8	8		REF1704
4	J1, J2, J3, J4		SPC4078	CP-102A-ND	DK	PJ-102A	CONN PWR JACK 2X5.5MM SOLDER	$0.60	65743868.xlsx	30	*		REF239
1	PWR		CHIP-LED0805	475-1410-1-ND	DK	LG R971-KN-1	LED GREEN DIFFUSED 0805 SMD	$0.17	74391652.xlsx	50	34		REF625
2	Q1, Q2	PNP	SOT223	1727-2371-1-ND	DK	PBHV9414ZX	TRANS PNP 140V 4A SOT223	$0.49	77747935.xlsx	25	*		REF1540
2	Q3, Q4	2N2222	SOT23-BEC	641-1942-1-ND	DK	MMBT2222A-G	TRANS NPN 40V 600MA SOT23	$0.13	78056136.xlsx	200	48		REF1341
1	R1	1200	R0805	P1.2KACT-ND	DK	ERJ-6GEYJ122V	RES SMD 1.2K OHM 5% 1/8W 0805	$0.10	NoOrderID.xlsx	927	927		REF1776
1	R2	50M	HB_RES	SM104FE-50M-ND	DK	SM104035005FE	RES 50M OHM 1% 1.5W RADIAL, No not install	$4.65	70553215.xlsx	20	12		REF1018
4	R3, R4, R5, R6	1000	M0805	P4.7KADCT-ND	DK	ERJ-P06J472V	RES SMD 4.7K OHM 5% 1/2W 0805	$0.13	NoOrderID.xlsx	58	58		REF1790
1	R7	10K	M0805	P4.7KADCT-ND	DK	ERJ-P06J472V	RES SMD 4.7K OHM 5% 1/2W 0805	$0.13	NoOrderID.xlsx	58	58		REF1790